Curriculum Vitæ

Jules Berry – Postdoctoral Researcher

Personal Information

Jules Berry.
Postdoctoral Researcher in Applied Mathematics at Université Paris-Saclay – CentraleSupélec, France.

Languages: French (native), English (fluent)
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Academic Positions

Since 11/2025 **Postdoctoral Researcher**, L2S – Université Paris-Saclay – CentraleSupélec, France.
Supervisors: Riccardo Bonalli (L2S – CNRS) and Laurent Pfeiffer (L2S – INRIA).
Research topic: Risk-averse stochatic optimal control.

Education

2022 – 2025 **Ph.D. in Mathematics and their Interactions**, IRMAR – Université de Rennes – INSA Rennes, France.
Thesis: Contributions to the study of second order mean field games: approximation of stable solutions and sticky processes on networks – tel-05393129.
Defense date: October 23, 2025 at INSA Rennes.
Advisors: Olivier Ley (IRMAR – INSA Rennes) and Francisco Silva (XLIM – Université de Limoges).

2020–2022 **Master in Fundamental Mathematics**, Université de Rennes, France, with Highest Honors.

Publications

[J4] Berry, J., Ley, O., & Silva, F. J. (2026). A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem via metric regularity techniques and applications to nonlinear PDEs. *Foundations of Computational Mathematics*.

[J3] Berry, J., & Colantoni, F. (2026). Sticky diffusions on star graphs: characterization and Itô formula. *Stochastic Processes and their Applications*.

[J2] Berry, J., & Camilli, F. (2025). Stationary Mean Field Games on networks with sticky transition conditions. *ESAIM: Control, Optimisation and Calculus of Variations*.

[J1] Berry, J., Ley, O., & Silva, F. J. (2025). Approximation and perturbations of stable solutions to a stationary mean field game system. *Journal de Mathématiques Pures et Appliquées*.

Preprints

[P2] Basti, A., Berry, J. and Camilli, F. (2026). Markov Chain Approximation of Sticky Diffusions and Hamilton-Jacobi-Bellman Equations on Networks, hal-05706139.

[P1] Berry, J. (2025). Some error estimates for semidiscrete finite element approximations of stable solutions to mean field game systems, hal-05365007.

Communications

Invited talks

Oct. 2026 **Hamilton-Jacobi Equations and Mean Field Games: from Modeling to Numerics via Analysis,** *Approximation of stable solutions to second order mean field game systems*, Pisa, Italy.

Sept. 2025 **Durham Symposium on Mean Field Games,** *Approximation of stable solutions to second order mean field game systems*, Durham, United Kingdom.

Oct. 2024 **ANR COSS Workshop,** *Sticky diffusion processes on networks and corresponding Mean Field Games*, Rennes, France.

Feb. 2024 **Rennes-Tours Workshop,** *Approximation of stable solutions to a stationary mean field game system*, Rennes, France.

Talks in seminars

Feb. 2026 **Séminiare EDP,** *Mean field games of sticky diffusions on networks*, Laboratoire de mathématiques de Besançon, Besançon, France.

Dec. 2024 **MOD Seminar,** *Sticky diffusion processes on networks and corresponding Mean Field Games*, XLIM, Limoges, France.

April 2024 **Seminario di Modellistica Differenziale Numerica,** *Approximation of stable solutions to second order mean field game systems*, La Sapienza Università di Roma, Rome, Italy.

Contributed talks

Jun. 2026 **Congrès National d'Analyse Numérique (CANUM),** *A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem*, Saint-Jacut-de-la-mer, France.

Mar. 2026 **Journées SMAI MODE,** *Approximation of stable solutions to second order mean field game systems*, Nice, France.

Mar. 2026 **Congrès des Jeunes Chercheurs en Mathématiques Appliquées,** *A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem*, Champs-sur-Marne, France.

Sep. 2025 **European Conference on Numerical Mathematics and Advanced Applications (ENUMATH),** *Approximation of stable solutions to second order mean field game systems*, Heidelberg, Germany.

Jan. 2024 **3rd International Conference on Variational Analysis and Optimization – In Honor of Boris Mordukhovich,** *A theorem of Brezzi, Rappaz, and Raviart from the point of view of variational analysis*, Santiago, Chile.

Jan. 2024 **Conference on Numerical methods for optimal transport problems, mean field games, and multi-agent dynamics,** *Approximation of stable solutions to a stationary MFG system*, Valparaiso, Chile.

Posters

Nov. 2025 **Première rencontre nationale du RT Optimisation,** *Mean Field Games on networks with sticky transition conditions*, Lyon, France.

Juin 2024 **Summer School on Machine Learning and Optimal Control,** *Approximation of non-differentiable nonlinear problems*, Gaeta, Italie.

Mars 2024 **Journées SMAI MODE,** *Approximation of stable solutions to a stationary mean field game system*, Lyon, France.

Teaching

2026 **Introduction to stochastic calculus**, ENSTA Paris-Saclay, Tutorial sessions (10h).
Topics: Brownian motion, stochastic integrals, stochastic differential equations, Girsanov's theorem.

2022–2025 **Analysis 3**, INSA Rennes, Tutorial sessions (48h/year), Undergraduate level (2nd year).
Topics: improper integrals, numerical series, power series, Fourier series, differential calculus.
Analysis Tools for Engineering, INSA Rennes, Tutorial sessions (10h/year), Undergraduate level (3rd year).
Topics: Lebesgue integrals, Fourier transform, complex analysis.

Academic Supervision

Master's Students

11/2025 – 4/2026 **Chaimae El Omari**, research initiation project, CentraleSupélec.
Co-supervised with Laurent Pfeiffer (L2S – INRIA).
Topic: Potential mean field games with non-differentiable Hamiltonians.

11/2024 – 12/2024 **Angelina Jammart**, Master's student seminar, Université de Rennes.
Co-supervised with Othmane Jerhaoui (IRMAR – INSA Rennes).
Topic: Introduction to viscosity solutions for Hamilton-Jacobi equations.

Funding

2025 **SMAI BOUM**, Funding for a week of collaborative research at CIRM.
Project: Comparison of macroscopic models for crowd motion.
In collaboration with Théo Girard (Univ. Tours) and Florian Peru (Univ. Franche-Comté).

Awards

2024 **Poster Award**: First prize ex-aequo, SMAI MODE Conference, Lyon.

Research Visits

Jul. 2026 **Université Marie-et-Louis-Pasteur**, Besançon, invited by Carlotta Donadello and Nathaël Alibaud, 3 days.

March–May 2024 **Università di Roma la Sapienza**, invited by Fabio Camilli, 3 months.

2022–2025 **Université de Limoges**, invited by Francisco Silva, 6 one-week visits.

Institutional Activities

2024–Present **Reviewer** for *J. Math. Anal. Appl.* (1), *NoDEA* (1), *Electron. J. Probab.* (1).

Last updated October 17, 2026